

Load-Aware A* Path Planning for Quadruped Robots

Day 19 of 30 · Week 3: Comprehensive Testing · Member 1 — Figures for Paper

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1. Review of methodology

III-A · Problem Formulation

1. Grid definition $G(r,c) \in \{0,1\}$ is clear and correctly stated (Eq. 1).
2. Path definition (Eq. 2) with 4-connected constraint is accurate and complete.
3. L defined ambiguously. "L = geometric path length (steps)" is stated after Eq. 2 as a standalone label. L should be formally defined inside the equation or as a numbered definition — e.g. $L = |\pi| - 1$ (number of edges, not nodes). As written it is easy to miscount. Recommend adding this inline.

III-B · Quadruped Robot Model

1. Foot contact coordinates (Eq. 3) are consistent with $h_s = 0.4$ m and a 1.0 m cell.
2. Eq. 3 — coordinate convention mismatch. You define positions as $(c \pm h_s, r \pm h_s)$, i.e. (col, row) = (x, y). But Eq. 5/6 and the rest of the paper use row-column indexing. You need one explicit statement — either "we use $(x,y) = (\text{col}, \text{row})$ " or define all foot points as $(\text{row} \pm h_s, \text{col} \pm h_s)$. Without this, any reader checking your CoM calculation against Eq. 4 will get confused about which axis is which.
3. CoM formula (Eq. 4) is correctly stated as weighted average of body and payload masses.
4. x_b not formally defined. You state " x_b = foot centroid" inline, but this should be an equation: $x_b = (p_{FL} + p_{FR} + p_{RL} + p_{RR}) / 4$. Reviewers checking your stability derivation will want this explicit.
5. Payload CoM position x_{payload} is underspecified. You state "payload CoM position" but never define how x_{payload} is set per cell (e.g. always at body centre, or at a fixed offset). Table I says "payload positions rows 3–7, cols 3–7" — clarify whether $x_{\text{payload}} = x_b$ or varies per scenario.

III-C · Standard A*

1. Evaluation function (Eq. 7) correctly defined as $f(n) = g(n) + h(n)$.
2. Manhattan heuristic (Eq. 8) is admissible and consistent for 4-connected grids — correctly justified. Verification $h((0,0),(9,9)) = 18$ is correct.
3. Uniform step cost (Eq. 9) is correctly stated. Optimality and completeness claims are accurate.

III-D · Load-Aware A*

1. Modified cost function (Eq. 10) is well-formed. Reducing to Standard A* when $\alpha = 0$ is correctly noted.
2. Admissibility claim needs justification. You state "h(n) is unchanged — admissibility preserved" but this is not automatically true when g(n) is inflated by $\alpha \cdot S(n)$. Admissibility requires $h(n) \leq h^*(n)$ where h^* is the true optimal cost under the new cost function. Since $\alpha \cdot S(n) \geq 0$, the modified path cost is always $\geq$ path length, so the true optimal modified cost $\geq$ path length $\geq h(n)$. The argument holds, but you must state it explicitly, not assume it. One sentence suffices.
3. S_map pre-computation (Eq. 12) and $O(1)$ lookup claim are correct. $O(\text{rows} \times \text{cols})$ build cost is accurate.
4. Novelty table N3 references Day 7 tuning — this is an internal note, not appropriate for a final submission. Replace with "selected via systematic tuning across $\alpha \in \{0.1, 0.5, 1.0, 1.5, 2.0\}$ over 10 scenarios" and remove the day reference.
5. The novelty table appears to be cut off. N3's last line ends mid-sentence: "Minimum a producing co...". This is either a PDF truncation or missing text — verify the full sentence is present in your source document.

II-E · Implementation Details

1. Data structure table (open_list, closed_set, came_from, g_score, S_map) is accurate, complete, and clearly presented.
2. Time complexity claims — $O(\log n)$ heap, $O(1)$ set/dict — are correct for Python's heapq and dict.
3. S_map is marked with an asterisk (*) in the table but no footnote is visible in the document. If the asterisk means "Load-Aware only", state that explicitly in the table (e.g. add a column "Used by" with "Both" or "LA only").
4. Python version and Colab runtime. "Python 3 (Google Colaboratory)" is fine but consider adding the specific Python version (e.g. 3.10) and whether GPU/TPU was used, as Colab runtimes vary. This aids reproducibility.

III-F · Experimental Setup

1. Inconsistency: 20 scenarios vs "80 runs". Table I justifies Grid size 10×10 as "80 runs feasible", but the Scenarios row says 20. $20 \text{ scenarios} \times 2 \text{ algorithms} = 40 \text{ runs}$. 80 runs would require 40 scenarios or 4 algorithms. Either the scenario count or the run count is wrong — clarify.
2. Metrics (L, C, N, T, S_avg, M_min, D, dL, dT%) are well-defined and complete. Formulas for dL and dT% are correctly stated.
3. Parameter table is well-structured with justifications. Stability threshold $M > 0.05$ m citing literature is good practice.
4. Payload range 3–9 kg (30–90% body mass) is correctly derived from $m_b = 10 \text{ kg}$ and the note "below 3 kg: no divergence" provides useful experimental context.
5. Seeding strategy not specified. You note "seeds are pre-validated as solvable" but don't list the actual seed values or describe how you validated them (e.g. ran BFS/A* and confirmed a path exists). Add a line or a supplementary table with seed values for full reproducibility.